

# TRAI TELECOM ANALYTICS

## End-to-End Data Analytics Portfolio Case Study

**TRAI PMR Data | July 2025 - July 2026**

**Excel / Power Query -> SQL Server -> Power BI**

22 Licensed Service Areas | 5 Service Providers | 4 Interactive Dashboard Pages

## 1. Project at a Glance

This project transforms TRAI Performance Monitoring Report (PMR) data into an analytical solution for evaluating telecom network performance, customer experience, geographic variation and reporting completeness. The project was built from source PDFs through data preparation, SQL modeling, validation and Power BI reporting.

| Reporting Period    | Service Providers | LSAs | Network Records |
|---------------------|-------------------|------|-----------------|
| Jul 2025 - Jul 2026 | 5                 | 22   | 1,144           |

## 2. Business Problem

Telecom performance is multi-dimensional: a provider can have strong network availability while differing materially on speed, latency, dropped calls, customer-care responsiveness or regional performance. Raw PMR reporting also contains missing values that can affect provider comparisons. The analytical challenge was therefore to create one traceable solution that answers both performance and data-quality questions.

## 3. Key Questions

- How do providers compare on availability, latency, packet drop, call quality and speed?
- How does customer-service performance vary across providers and months?
- Which Licensed Service Areas show stronger or weaker network performance?
- Are important KPIs consistently reported, and where do reporting gaps exist?
- What findings are important enough to support management or operational action?

## 4. End-to-End Workflow

1. Collected and archived TRAI PMR source PDFs.
2. Extracted and consolidated PDF tables in Excel using Power Query.
3. Validated missing rows, NA values, provider/LSA dimensions and reporting months.
4. Loaded cleaned datasets into SQL Server and created analytical views.
5. Built Power BI measures, relationships, KPI benchmarks and four dashboard pages.
6. Investigated provider, monthly, LSA and data-completeness patterns.
7. Documented findings, limitations and business recommendations.

## 5. Data & Technical Architecture

| Layer         | Tool                | Role                                               | Output                                    |
|---------------|---------------------|----------------------------------------------------|-------------------------------------------|
| Source        | TRAI PMR PDFs       | Raw telecom performance reporting                  | Archived source files                     |
| Preparation   | Excel + Power Query | Extraction, consolidation, cleaning and validation | Final analytical workbook                 |
| Data Layer    | SQL Server / SSMS   | Structured tables and analytical views             | Provider, monthly, LSA and coverage views |
| Visualization | Power BI            | DAX, relationships, KPI analysis and dashboards    | 4-page interactive report                 |

## 6. Power BI Solution

### Dashboard 1 - Network Performance & Service Quality

Monthly provider analysis covering network availability, typical/P80 speed, latency, packet drop, dropped-call rates and customer-service accessibility.

### Dashboard 2 - Customer Experience & Service Quality

Billing complaints, complaint trends, call connection quality and customer-care responsiveness.

### Dashboard 3 - LSA & Geographic Performance

22-LSA comparison plus geographic network-quality mapping, with packet drop used to create meaningful bubble-size variation.

### Dashboard 4 - Data Quality & Reporting Completeness

Provider and monthly reporting completeness, unreported network records and overall metric reporting status.

## 7. Selected KPI Benchmarks

| Metric                       | Reference | Interpretation   |
|------------------------------|-----------|------------------|
| Network Availability         | >= 99%    | Higher is better |
| Latency                      | <= 75 ms  | Lower is better  |
| Packet Drop                  | <= 3%     | Lower is better  |
| DCR-CS / DCR-PS              | <= 2%     | Lower is better  |
| Customer Care Accessibility  | >= 95%    | Higher is better |
| Calls Answered within 90 sec | >= 95%    | Higher is better |

## 8. Key Findings

**Network quality improved:** Average latency improved from 31.69 ms in July 2025 to 26.63 ms in July 2026, while average packet drop declined from 0.76% to 0.50%.

**Availability remained consistently high:** Average network availability stayed around 99.9% across the reporting period.

**RJIL led download-speed performance:** Average typical download speed was 28.21 Mbps for RJIL, followed by AIRTEL at 18.34 Mbps and VIL at 15.00 Mbps.

**Customer service was generally strong:** VIL recorded the highest average calls answered within 90 seconds at 99.90%; BSNL was lowest at 95.67%, still above the 95% aggregate benchmark.

**BSNL showed a consistency opportunity:** Of 18 individual observations below the 95% call-answering benchmark, 15 were BSNL observations.

**Regional differences were material:** NE, AS and WB showed the highest average latency, while BR, UPE and GJ were among the strongest LSAs for typical download speed.

**Reporting completeness affected comparison:** DCR-CS was missing for all 286 RJIL network records, reducing overall DCR-CS reporting completeness to 75%. MTNL also had missing DCR-PS, latency and packet-drop values.

## 9. Data Quality Investigation

A dedicated data-quality layer was included because missing values can create misleading provider comparisons. The project used an Issue -> Detection -> Investigation -> Decision -> Correction -> Validation approach.

| Issue               | Investigation                                          | Decision                         | Outcome                    |
|---------------------|--------------------------------------------------------|----------------------------------|----------------------------|
| PDF extraction gaps | Source pages and row counts were rechecked             | Restore omitted rows from source | Final workbook reconciled  |
| NA / blank values   | Confirmed as unavailable/not-reported rather than zero | Preserve as null                 | Averages not distorted     |
| RJIL DCR-CS gap     | All 286 missing DCR-CS records traced to RJIL          | Do not impute                    | Gap exposed in Dashboard 4 |
| MTNL sparse metrics | Missing DCR-PS, latency and packet drop identified     | Do not fabricate values          | Coverage shown separately  |

## 10. Business Recommendations

- Prioritize drill-down analysis for higher-latency LSAs such as NE, AS and WB to isolate recurring provider/month patterns.
- Review BSNL customer-care queue, staffing and peak-demand drivers behind below-benchmark 90-second answering observations.
- Use provider speed differences to investigate capacity, tariff mix and service-plan composition rather than relying only on national averages.
- Treat reporting completeness as a governance KPI before direct provider benchmarking.
- Resolve or formally explain recurring source-reporting gaps instead of imputing telecom KPI values.
- Use Power BI for monitoring and drill-down while retaining Excel/Power Query and SQL as the auditable preparation and transformation trail.

## 11. Skills Demonstrated

|                                                   |                                                    |
|---------------------------------------------------|----------------------------------------------------|
| Data acquisition from semi-structured PDF sources | Power Query extraction, consolidation and cleaning |
| Excel-based validation and reconciliation         | SQL Server data modeling and analytical views      |
| Data-quality investigation and null handling      | Power BI relationships and DAX measures            |
| KPI benchmarking and variance logic               | Time-series, provider and geographic analysis      |
| Executive dashboard design                        | Business insight and recommendation writing        |

## 12. Project Limitations

- The analysis covers July 2025 through July 2026 and is not a complete historical view.
- Missing source values were preserved; no KPI values were estimated or imputed.
- Provider and LSA averages can hide month-level variation and should be paired with dashboard drill-down.
- Typical and P80 speed measures are treated descriptively where no formal project threshold was used.
- The archived telecom-subscription dataset was excluded because it could not be integrated reliably into the final model.

## 13. Portfolio Summary

This project demonstrates a complete analytics workflow from raw public-sector telecom reporting to a validated, decision-oriented BI solution. Its strongest portfolio value is not a single dashboard visual; it is the traceable process connecting source-data challenges, cleaning decisions, SQL modeling, KPI logic, geographic analysis, data-quality investigation and business recommendations.

**PROJECT STATUS: COMPLETE**